

Eigenvalues and inverse problems

Two permutation matrices generate the doubly stochastic spectral boundary

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging reflects a global spectral-boundary description using a restricted combinatorial family; community impact connects stochastic matrices, inverse spectra, and convex geometry.

Problem statement

Let $\mathcal{D}_n$ consist of the real entrywise-nonnegative $n \times n$ matrices whose row and column sums are all one, and define the single-eigenvalue region

$$D_n = \{z \in \mathbb{C} : z \in \sigma(A) \text{ for some } A \in \mathcal{D}_n\}.$$

Prove or disprove that, for every integer $n \geq 1$ and every $z \in \partial D_n$, there exist $n \times n$ permutation matrices P, Q and $t \in [0, 1]$ such that

$$\det(zI - tP - (1 - t)Q) = 0.$$

The boundary is taken in the usual topology of $\mathbb{C}$, and $P = Q$ is allowed. The task concerns one eigenvalue, not simultaneous realization of a prescribed full spectrum. It would reduce boundary computation to a finite collection of one-parameter matrix families. Counterexamples for convex hulls of other matrix groups do not automatically apply to all permutation matrices.

References

Harlev, Johnson, and Lim, *The Doubly Stochastic Single Eigenvalue Problem: A Computational Approach*, Conjecture 2.7 and §6; published in *Experimental Mathematics* 31 (2022), 936–945. Verbeke and Ginis, [ILAS 2026 abstract](#), p. 150.

Earlier status check — 2026-09-08

Searches for *Harlev Johnson Lim boundary conjecture* and *doubly stochastic boundary conjecture 2026* found the 2026 authors reporting numerical support through order 25, rather than a proof. The older Perfect–Mirsky conjecture fails at order five; that failure does not refute this different assertion.

Audit update — 2026-09-10

Rechecked the [Boundary Conjecture source](#) and the [ILAS 2026 abstracts](#), p. 150. The latter reports further computational exploration, not a general proof. Searches for subsequent doubly stochastic spectral-boundary results found no resolution; the refuted Perfect–Mirsky conjecture is a different claim.